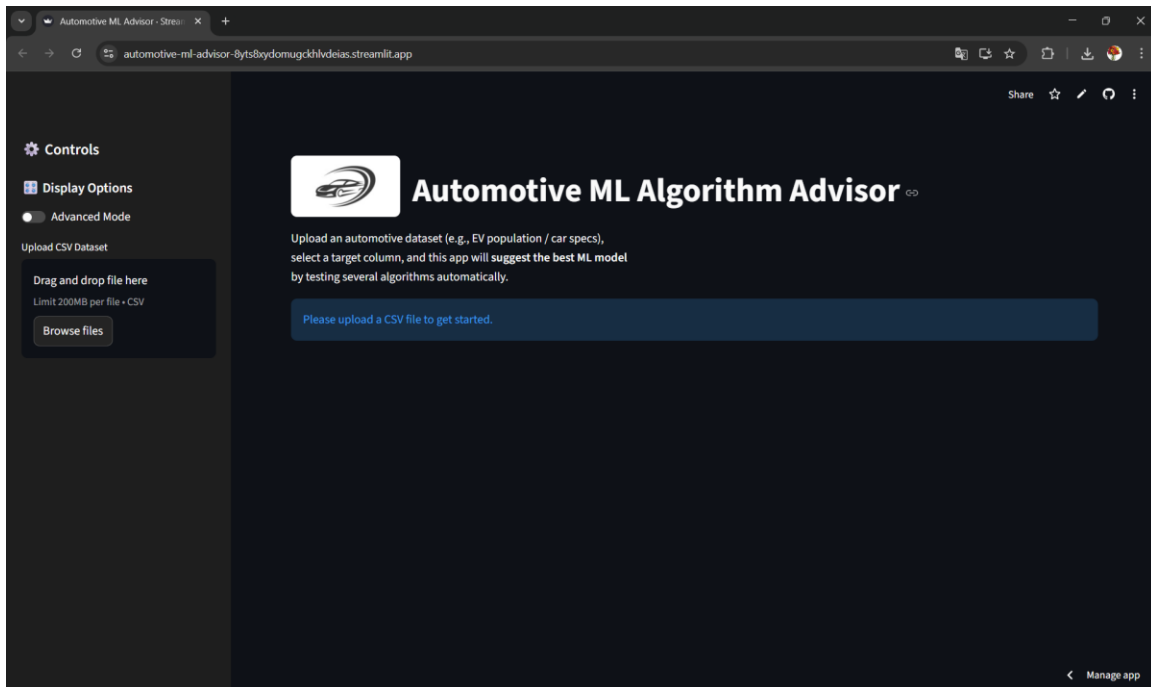


# Automotive ML Advisor – GUI Documentation

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## Screen 1



Home screen of the Automotive ML Algorithm Advisor showing application title, upload section, and sidebar controls. This page acts as the starting interface where users upload datasets and configure settings before running analysis.

## Screen 2

The screenshot shows the 'Automotive ML Algorithm Advisor' web application. The left sidebar contains controls for display options (Advanced Mode), CSV upload (49.0MB), target column selection ('Electric Range'), problem type selection (Regression), and model selection mode (Automatic/Manual). The main area displays a 'Preview of Data' with a table of 10 rows and 11 columns. The columns are VIN (1-10), County, City, State, Postal Code, Model Year, Make, Model, Electric Vehicle Type, and Clean Alternative Fuel Ve. The table shows various vehicle models like BMW X5, Tesla Model 3, and Ford Fusion. Below the table, it states 'Original Shape: 210165 rows x 17 columns' and has a 'Manage app' link.

|   | VIN (1-10) | County    | City      | State | Postal Code | Model Year | Make    | Model    | Electric Vehicle Type                  | Clean Alternative Fuel Ve  |
|---|------------|-----------|-----------|-------|-------------|------------|---------|----------|----------------------------------------|----------------------------|
| 0 | SUXTA6COXM | Kitsap    | Seabeck   | WA    | 98380       | 2021       | BMW     | X5       | Plug-in Hybrid Electric Vehicle (PHEV) | Clean Alternative Fuel Ve  |
| 1 | SU3E1EB1J  | Kitsap    | Poulsbo   | WA    | 98370       | 2018       | TESLA   | MODEL 3  | Battery Electric Vehicle (BEV)         | Clean Alternative Fuel Ve  |
| 2 | WP0AD2A73G | Snohomish | Bothell   | WA    | 98012       | 2016       | PORSCHE | PANAMERA | Plug-in Hybrid Electric Vehicle (PHEV) | Not eligible due to low bi |
| 3 | SU3E1EB5J  | Kitsap    | Bremerton | WA    | 98310       | 2018       | TESLA   | MODEL 3  | Battery Electric Vehicle (BEV)         | Clean Alternative Fuel Ve  |
| 4 | 1N4A21CP3K | King      | Redmond   | WA    | 98052       | 2019       | NISSAN  | LEAF     | Battery Electric Vehicle (BEV)         | Clean Alternative Fuel Ve  |
| 5 | 3FA6P0PUSF | Snohomish | Bothell   | WA    | 98012       | 2015       | FORD    | FUSION   | Plug-in Hybrid Electric Vehicle (PHEV) | Not eligible due to low bi |
| 6 | 5VJGDDEXL  | King      | Renton    | WA    | 98055       | 2020       | TESLA   | MODEL Y  | Battery Electric Vehicle (BEV)         | Clean Alternative Fuel Ve  |
| 7 | SUXTS1C06M | King      | Seattle   | WA    | 98107       | 2021       | BMW     | X3       | Plug-in Hybrid Electric Vehicle (PHEV) | Not eligible due to low bi |
| 8 | 1N4A20CP0F | King      | Bellevue  | WA    | 98007       | 2015       | NISSAN  | LEAF     | Battery Electric Vehicle (BEV)         | Clean Alternative Fuel Ve  |
| 9 | SU5A1E20H  | King      | Seattle   | WA    | 98125       | 2017       | TESLA   | MODEL S  | Battery Electric Vehicle (BEV)         | Clean Alternative Fuel Ve  |

Dataset preview interface displaying uploaded CSV contents with rows, columns, and selected target column. Helps users verify dataset correctness before training models.

## Screen 3

The screenshot shows the 'Dataset Summary' and 'Dataset Validation' panels. The 'Dataset Summary' panel displays 'Original Shape: 210165 rows x 17 columns', 'Rows: 3000', 'Columns: 17', and 'Missing Values: 1'. It also shows 'Training Shape (after sampling): 3000 rows x 17 columns' and 'Auto-detected problem type: Regression based on target column.' The 'Dataset Validation' panel shows a green message: 'Dataset looks suitable for ML modeling.' and a 'Run Model Suggestion' button. The left sidebar shows the same controls as Screen 2, with 'Model Selection Mode' set to 'Automatic (Recommended)'.

Original Shape: 210165 rows x 17 columns

Rows: 3000, Columns: 17, Missing Values: 1

Training Shape (after sampling): 3000 rows x 17 columns

Auto-detected problem type: Regression based on target column.

Time-series dataset detected using column: Model Year

Using 5 numeric and 8 categorical feature columns.

Time-series detected — disabling SMOTE & stratified split

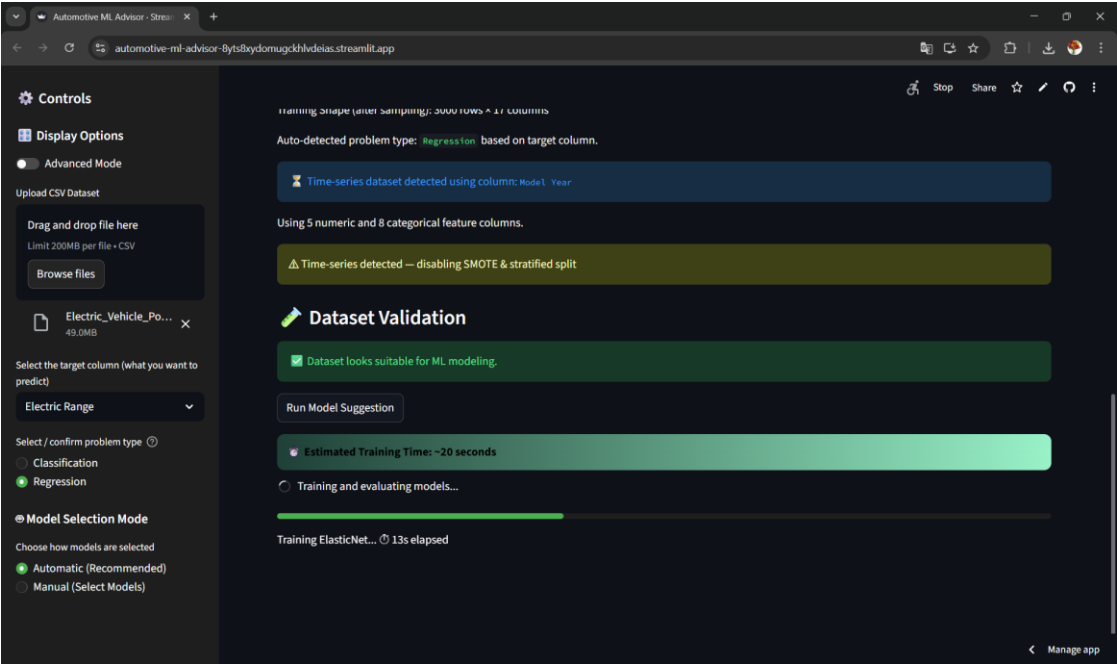
Dataset Validation

Dataset looks suitable for ML modeling.

Run Model Suggestion

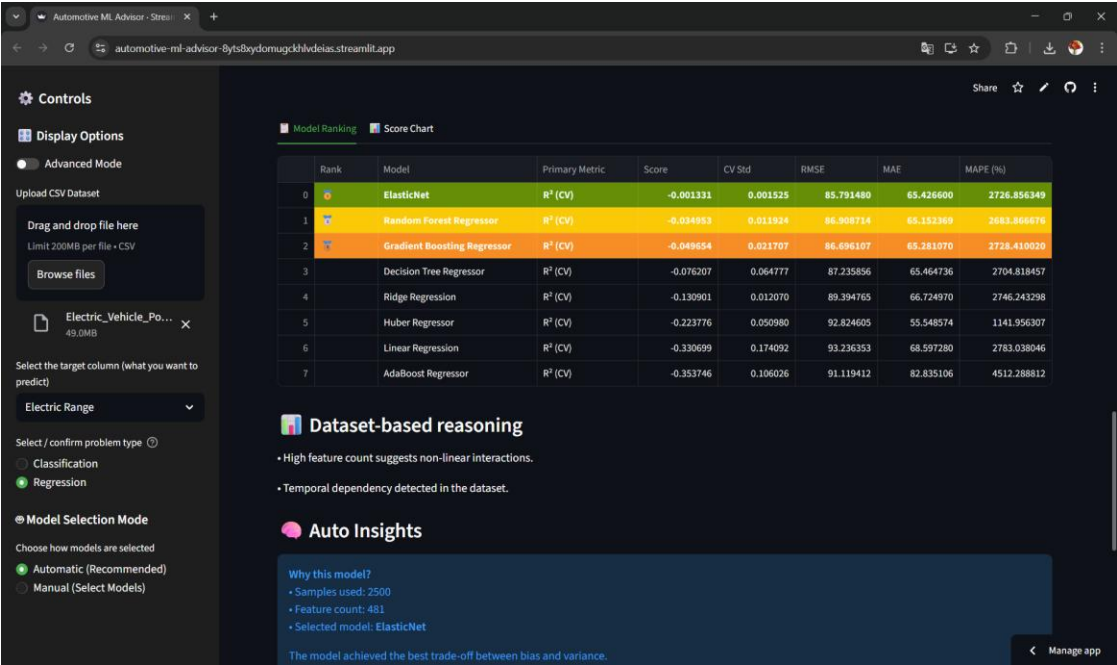
Dataset summary panel presenting dataset size, column count, missing values, and detected problem type. This ensures users understand dataset structure before modeling. In control panel, Model selection mode (Automatic & Manual) is also available

Screen 4



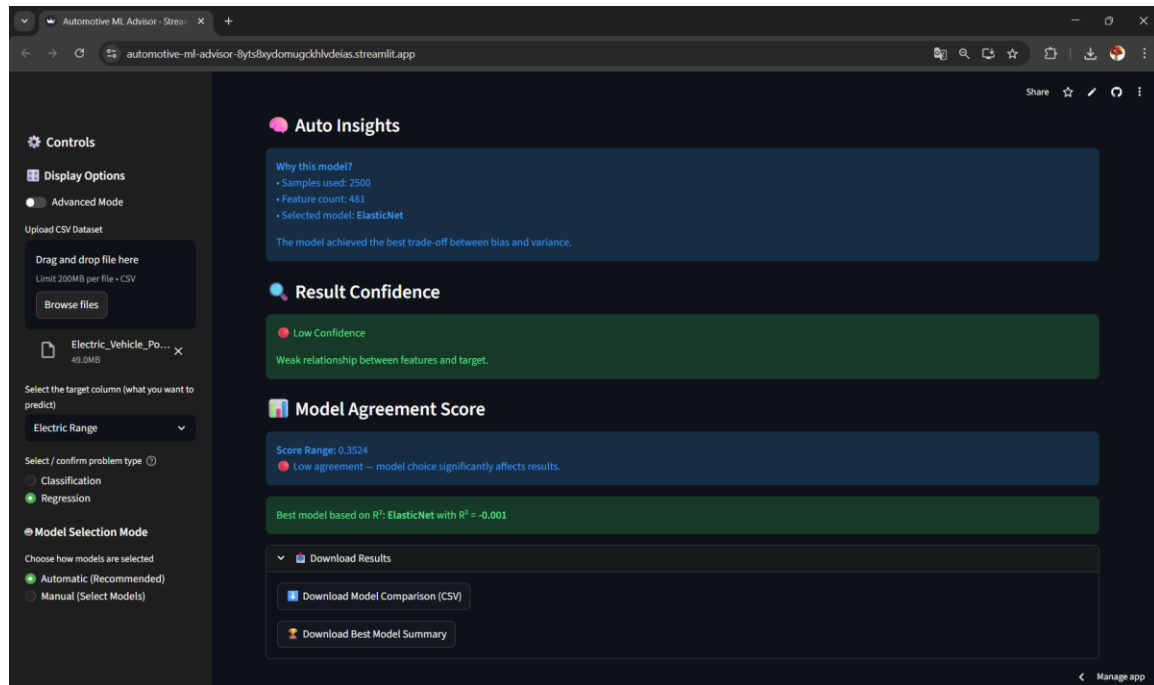
Advanced dataset validation screen indicating detected time-series column and feature distribution. System automatically warns about constraints like SMOTE limitations.

Screen 5



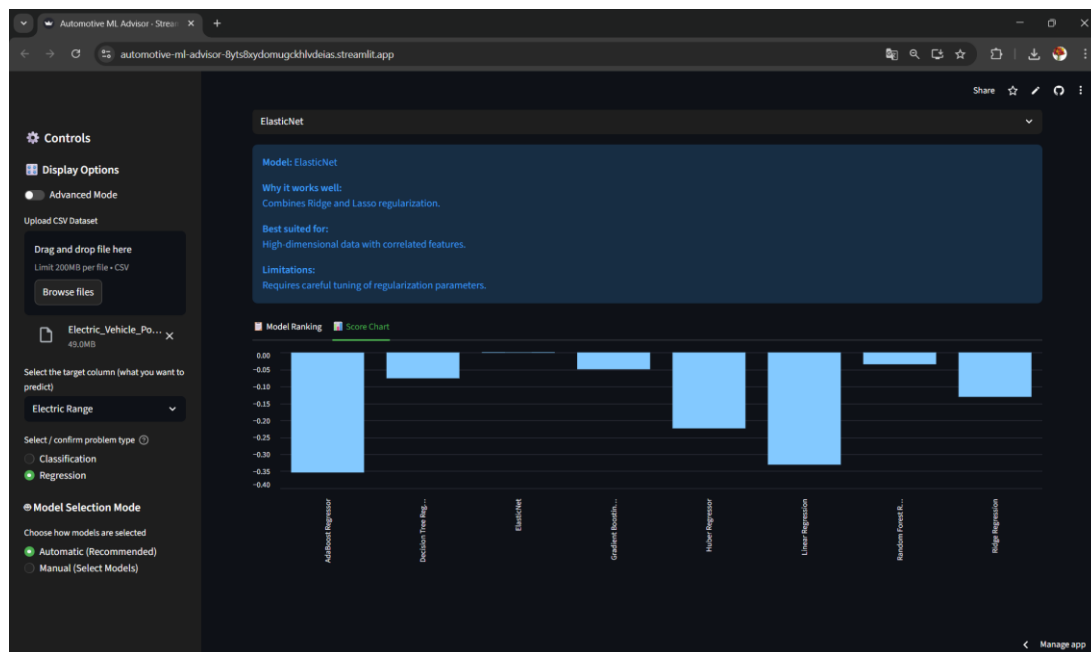
Model training screen showing progress bar and estimated training time. This visual feedback keeps users informed during model execution.

## Screen 6



Regression model ranking table comparing multiple algorithms using  $R^2$ , RMSE, MAE, and MAPE metrics. Helps identify best performing regression model.

## Screen 7



Auto insights section explaining why a particular model was selected based on dataset characteristics such as feature count and sample size.

## Screen 8

The screenshot displays the 'Compare Two Models' section of the Automotive ML Advisor. The interface is dark-themed with a sidebar on the left containing controls and display options. The main area shows a comparison between ElasticNet and Random Forest Regressor models. The ElasticNet model is selected for explanation, and its performance metrics are displayed in a table.

|                | 0                      | 1                    |
|----------------|------------------------|----------------------|
| Primary Metric | $R^2$ (CV)             | $R^2$ (CV)           |
| Score          | -0.0013307182046112809 | -0.03495316317848687 |
| CV Std         | 0.0015249594112643732  | 0.011923577415198991 |
| RMSE           | 85.79147977670951      | 86.90871393696354    |
| MAE            | 65.426600195334        | 65.15236911294735    |
| MAPE (%)       | 2726.856349254397      | 2683.8666764057807   |

Below the table, the 'Why this model fits your data' section provides an explanation for the ElasticNet model's selection. It states that the model works well because it combines Ridge and Lasso regularization, is best suited for high-dimensional data with correlated features, and has limitations requiring careful tuning of regularization parameters.

Result confidence panel indicating reliability of predictions. Provides interpretation of performance strength and warns if relationships are weak.

## Screen 9

The screenshot displays the 'Dataset Validation' section of the Automotive ML Advisor. The interface is dark-themed with a sidebar on the left containing controls and display options. The main area shows the results of dataset validation and model suggestion. The dataset is identified as 'Clean Alternative Fuel Vehicle (CAFE) Eligibility' and is suitable for ML modeling. The estimated training time is approximately 25 seconds.

Auto detected problem type: **Classification** based on target column.

CAFE like column detected. Converted to binary classes: 1 = eligible, 0 = not eligible/other.

Time-series dataset detected using column: **Model\_Year**

Using 6 numeric and 7 categorical feature columns.

Time-series detected -- disabling SMOTE & stratified split

Class distribution:

| Clean Alternative Fuel Vehicle (CAFE) Eligibility | count |
|---------------------------------------------------|-------|
| 0                                                 | 1988  |
| 1                                                 | 1012  |

SMOTE not applied (too few samples in a class). Using original class distribution. SMOTE disabled for time-series data to preserve temporal order.

**Dataset Validation**

Dataset looks suitable for ML modeling.

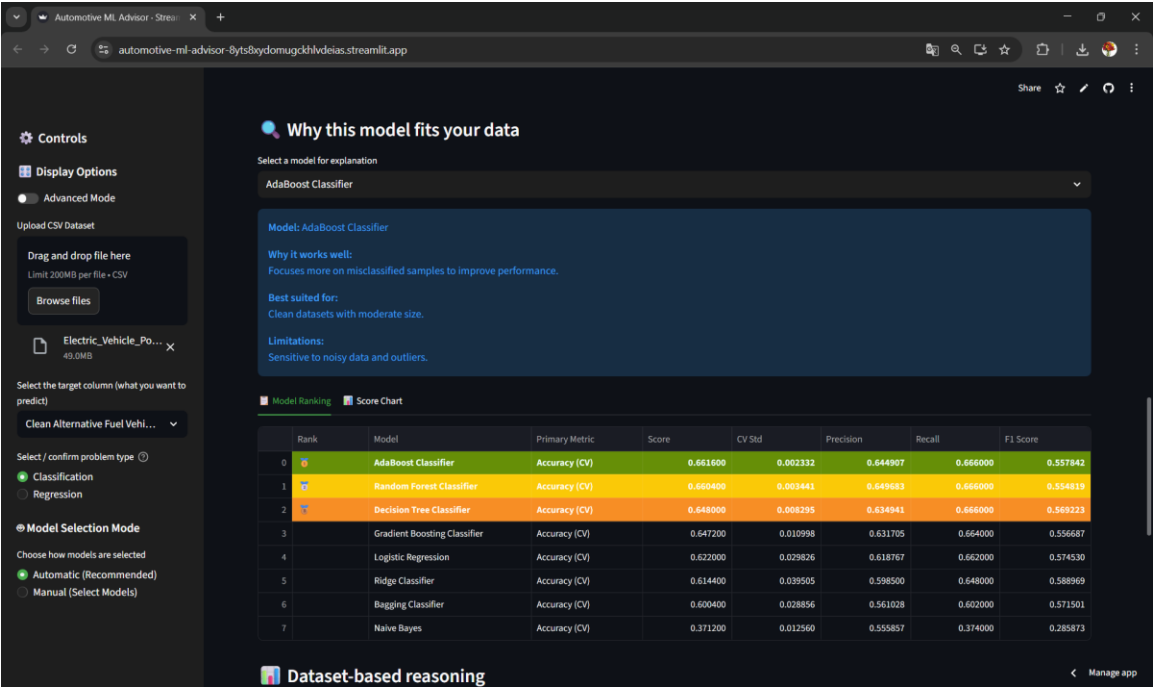
Run Model Suggestion

Estimated Training Time: ~25 seconds

Training completed

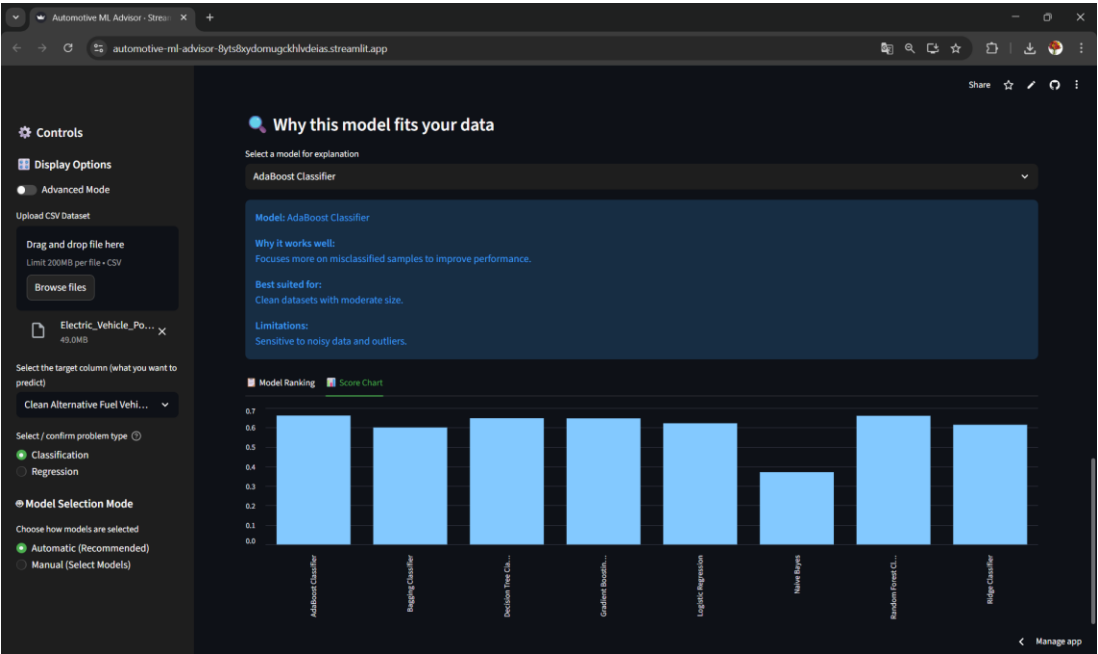
Model explanation panel describing selected algorithm logic, strengths, best use cases, and limitations. Enables interpretability for non-experts.

Screen 10



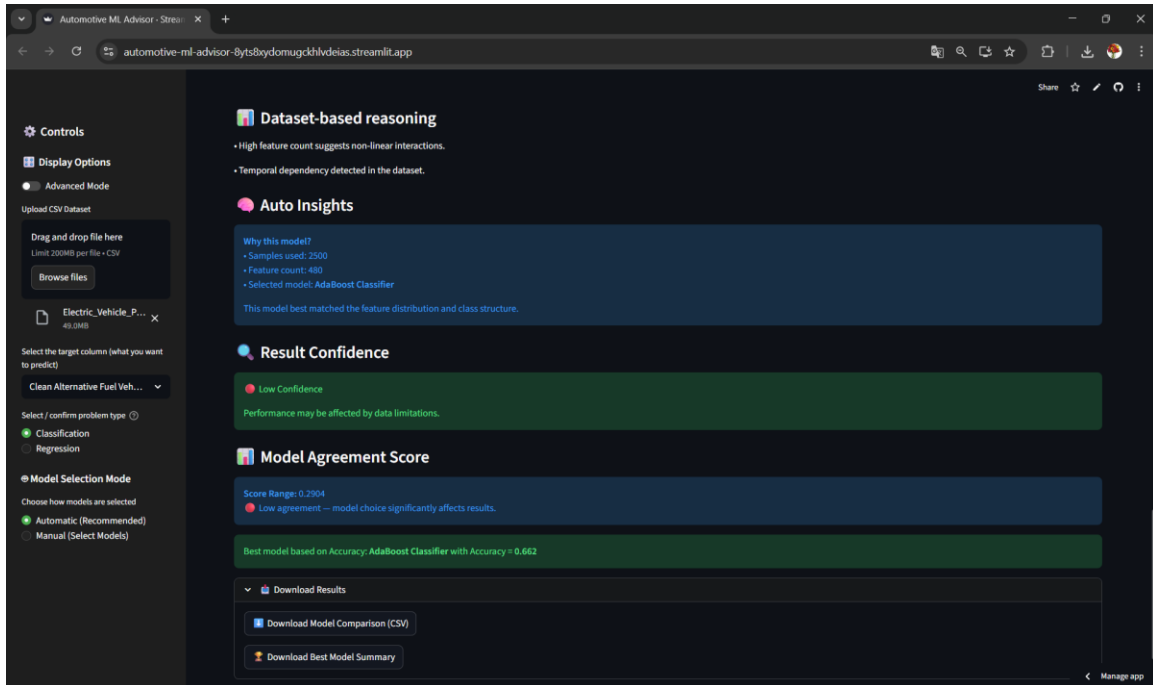
Model comparison bar chart visualizing performance scores across multiple algorithms for quick comparison.

Screen 11



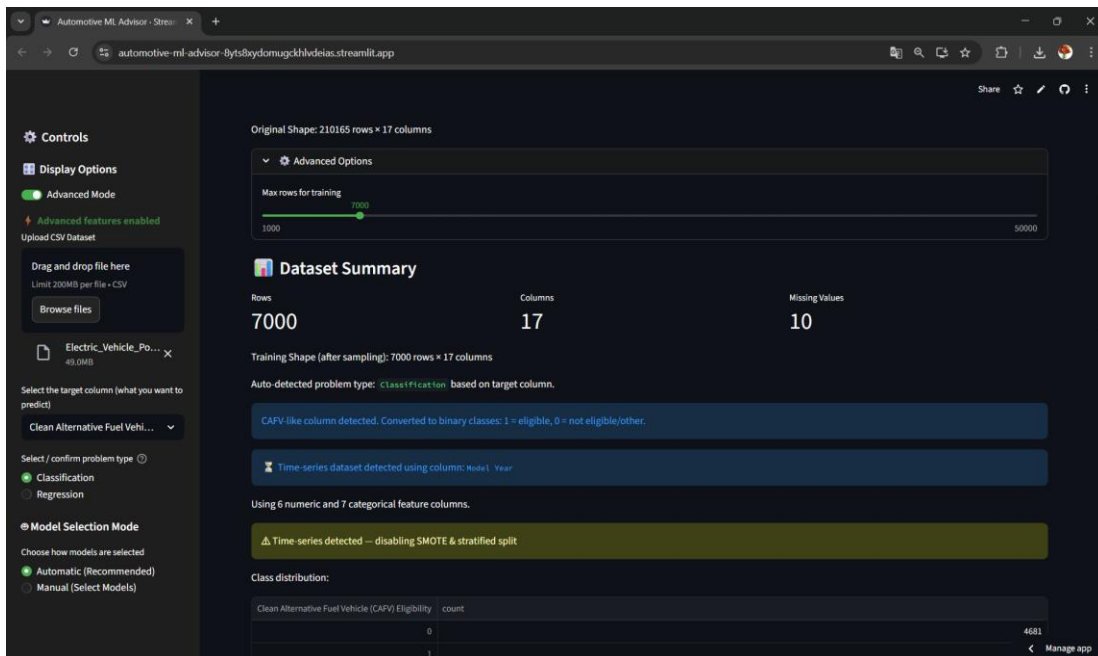
Dual model comparison view showing detailed metric comparison between two selected models. Helps users choose between alternatives.

## Screen 12



Classification dataset summary showing class distribution, detected task type, and preprocessing decisions taken automatically by system.

## Screen 13



Training completion screen confirming successful model evaluation and readiness of results for download.

## Screen 14

The screenshot shows the 'Automotive ML Advisor' web application. On the left, there's a sidebar with options to upload a CSV dataset, select a target column ('Clean Alternative Fuel Vehicle'), and choose a problem type (Classification or Regression). Below this, there's a 'Model Selection Mode' section with 'Automatic (Recommended)' and 'Manual (Select Models)' options. The 'Manual' mode shows a list of models to run: Logistic Regression, Ridge Classifier, Naive Bayes, and Decision Tree Classifier. The main panel displays a dataset preview with columns like VIN, Year, Make, Model, and Fuel Type. Below the preview, there's a 'Dataset Summary' section showing 7000 rows and 17 columns. It also indicates the training shape (7000 rows x 17 columns) and the auto-detected problem type (Classification). A note mentions 'CAFV like column detected. Converted to binary classes: 1 = eligible, 0 = not eligible/other.'

| VIN        | Year | Make   | Model   | Fuel Type                              | Fuel Type Description                   |
|------------|------|--------|---------|----------------------------------------|-----------------------------------------|
| 5KJYGEEXL  | 2020 | TESLA  | MODEL Y | Battery Electric Vehicle (BEV)         | Clean Alternative Fuel Vehicle Eligible |
| 5UXTSC06M  | 2021 | BMW    | X3      | Plug-in Hybrid Electric Vehicle (PHEV) | Not eligible due to low battery range   |
| 1N4AZ0CP0F | 2015 | NISSAN | LEAF    | Battery Electric Vehicle (BEV)         | Clean Alternative Fuel Vehicle Eligible |
| 5YJSALE20H | 2017 | TESLA  | MODEL S | Battery Electric Vehicle (BEV)         | Clean Alternative Fuel Vehicle Eligible |

Classification model ranking table comparing algorithms based on accuracy, precision, recall, and F1 score.

## Screen 15

The screenshot shows the 'Automotive ML Advisor' web application. The main panel displays a 'Why this model fits your data' section for the 'AdaBoost Classifier'. It explains why the model works well (focuses on misclassified samples) and lists its limitations (sensitive to noisy data and outliers). Below this, there's a 'Model Ranking' table comparing six models based on Accuracy (CV), Score, CV Std, Precision, Recall, and F1 Score. The 'Random Forest Classifier' is the top-ranked model.

| Rank | Model                        | Primary Metric | Score    | CV Std   | Precision | Recall   | F1 Score |
|------|------------------------------|----------------|----------|----------|-----------|----------|----------|
| 0    | Random Forest Classifier     | Accuracy (CV)  | 0.660440 | 0.014036 | 0.453255  | 0.673242 | 0.541768 |
| 1    | AdaBoost Classifier          | Accuracy (CV)  | 0.658384 | 0.018975 | 0.780402  | 0.674099 | 0.541797 |
| 2    | Decision Tree Classifier     | Accuracy (CV)  | 0.649983 | 0.012304 | 0.547468  | 0.862093 | 0.588363 |
| 3    | Logistic Regression          | Accuracy (CV)  | 0.642271 | 0.024065 | 0.555680  | 0.650086 | 0.560086 |
| 4    | Gradient Boosting Classifier | Accuracy (CV)  | 0.635417 | 0.034852 | 0.562197  | 0.671527 | 0.544048 |
| 5    | Bagging Classifier           | Accuracy (CV)  | 0.599591 | 0.008031 | 0.546590  | 0.457976 | 0.470352 |
| 6    | Naive Bayes                  | Accuracy (CV)  | 0.355842 | 0.009628 | 0.584965  | 0.369640 | 0.464400 |

Final insights and recommendation panel highlighting best model, agreement score, and downloadable results for reporting or reuse.